

Dr. Doc

TUTORIAL FOR MEDICINE STUDENTS

ANATOMY & GENERAL EMBRYOLOGY

Comprehensive 2013 ITM Exam Review Guide

A Question-by-Question Detailed Study Manual with Answers, Explanations, and Distracter Analysis

This review guide has been meticulously prepared and curated by **Dr. Doc Tutorial for Medicine Students**. It provides the full set of 50 multiple-choice questions from the 2013 Anatomy and General Embryology Preclerkship ITM Examination. Each question is fully transcribed and paired with its scientifically accurate correct answer, alongside a comprehensive, source-grounded explanation.

Every explanation details why the correct choice is biologically and anatomically sound, while breaking down the distracter choices to clarify why they are incorrect. This structured approach serves as an invaluable resource for medical students to reinforce key concepts in gross anatomy, histology, and general embryology.

STUDY SMART WITH DR. DOC TUTORIALS

Developing clinical competence through anatomical precision.

2013 ANATOMY ITM EXAM REVIEW

Detailed Solutions, Correct Answers, and Distracter Analysis

Question 1: Which of the following is NOT the concept of anatomy?

- A. This is the study of the human body structure on which the function lies.
- B. It is the study of the relationship of one structure to other.
- C. Physical examination is the clinical application of surface anatomy.
- D. None of the above

Correct Answer: D. None of the above

Explanation: Anatomy is defined as the basic medical science that studies the structure of the human body and the physical relationships between its various constituent parts [69, 100]. Surface anatomy provides knowledge of what structures lie under the skin and are perceptible to touch (palpable) in the living body, and physical examination is indeed its direct clinical application [73]. Therefore, statements A, B, and C are all correct concepts of anatomy, making 'None of the above' (D) the correct choice.

Question 2: Which of the following is NOT true about anatomical position?

- A. Palms of the subject stand erect facing forwards.
- B. The upper extremities are placed at the sides.
- C. The palms of the hands are turned forward.
- D. The head and eyes are facing forwards.
- E. None of the above

Correct Answer: E. None of the above

Explanation: In the standard anatomical position, the individual stands erect (upright), eyes and head facing directly forward, upper limbs placed at the sides with palms facing forward, and lower limbs parallel with feet flat on the floor and toes pointed forward [79, 109]. Since statements A, B, C, and D are all accurate descriptions of the anatomical position, none of them is false, making 'None of the above' (E) the correct answer.

Question 3: The plane that divides the body into superior and inferior parts:

- A. Mid-sagittal section
- B. Coronal section
- C. Transverse section
- D. Oblique section

Correct Answer: C. Transverse section

Explanation: The transverse plane (also called horizontal or axial plane) runs horizontally at a right angle to the vertical planes, dividing the body or organ into superior (upper) and inferior (lower) portions [81, 114]. In contrast, a mid-sagittal section (A) divides the body into equal right and left halves [80, 113], a coronal section (B) divides it into anterior and posterior portions [80, 113], and an oblique section (D) passes through at an angle [118].

Question 4: Assume that a patient is scheduled for thyroid surgery. In which one of the following positions should you place the patient in the operation room?

- A. Flexion
- B. Left lateral
- C. Prone
- D. Supine

Correct Answer: D. Supine

Explanation: For thyroid surgery, which involves an incision and exposure of the anterior neck, the patient must be placed in the supine position (lying on the back, face up) to allow optimal surgical access to the neck region [79, 109]. Plentiful extension of the neck is often added. Prone (C) is face-down [80, 110], left lateral (B) is lying on the left side, and flexion (A) is a joint movement, not a whole-body position [86, 124].

Question 5: Langerhans cells are located in:

- A. Stratum basale
- B. Stratum lucidum
- C. Stratum corneum
- D. Stratum spinosum
- E. Stratum granulosum

Correct Answer: D. Stratum spinosum

Explanation: Langerhans cells are dendritic, antigen-presenting immune cells of the skin. They are characteristically located in the stratum spinosum (prickly layer) of the epidermis, where they play a key role in cutaneous immunity. While they have cytoplasmic processes extending to other layers, their cell bodies reside in the stratum spinosum. Stratum basale (A) houses melanocytes and tactile cells; stratum corneum (C) is composed of dead, keratinized cells; stratum granulosum (E) is the granular layer; and stratum lucidum (B) is found only in thick skin.

Question 6: Fingerprint patterns are established due to:

- A. Stratum corneum
- B. Dermal papillary layer
- C. Stratum basale
- D. Dermal reticular layer
- E. Hypodermis

Correct Answer: B. Dermal papillary layer

Explanation: Fingerprints (epidermal ridges) are established by the dermal papillae, which are finger-like projections of the papillary layer of the dermis (the superficial layer of the dermis). These papillae project into the overlying epidermis, creating unique ridges on the skin surface of fingers and toes. The stratum corneum (A) and stratum basale (C) are epidermal layers, while the reticular layer (D) is the deeper, denser layer of the dermis, and the hypodermis (E) is the subcutaneous fat layer.

Question 7: What area of the skin structure is/are responsible for 'goose bump' formation of the skin?

- A. Pacinian corpuscle
- B. Arrector pili muscle
- C. Meissner's corpuscle
- D. Sensory nerve fiber
- E. None

Correct Answer: B. Arrector pili muscle

Explanation: The arrector pili muscle is a small band of smooth muscle fibers attached to the hair follicle and the dermal papillary layer. Under sympathetic stimulation (due to cold or emotional stress), these muscles contract, pulling the hair follicles upright and depressing the surrounding skin, which produces the characteristic skin elevations known as 'goose bumps' [7]. Pacinian (A) and Meissner's (C) corpuscles are encapsulated touch/vibration receptors.

Question 8: All are capsulated sensory receptors in the skin except?

- A. Hair follicle receptors
- B. Pacinian corpuscles
- C. Ruffini ending
- D. Meissner's corpuscles

Correct Answer: A. Hair follicle receptors

Explanation: Hair follicle receptors are unencapsulated (free) nerve endings that wrap around the base of hair follicles, acting as highly sensitive, rapidly adapting mechanoreceptors. In contrast, Pacinian corpuscles (B, deep pressure/vibration), Ruffini endings (C, skin stretch), and Meissner's corpuscles (D, light touch) are all structurally surrounded by a connective tissue capsule.

Question 9: Which of the following is/are true statement?

- A. Epiphysis are end of long bone that ossify from primary center of ossification.
- B. Diaphysis are elongated, cylindrical shaft of long bone that ossify from secondary center of ossification.
- C. Epiphysis are most active site of bone formation in developing bone.
- D. Epiphyseal line is responsible for growth and developing bone in length.
- E. All
- 6. None

Correct Answer: 6. None

Explanation: Let's review the statements: (A) is false because the epiphysis ossifies from secondary centers [13]; (B) is false because the diaphysis ossifies from the primary center [13]; (C) is false because the most active site of bone formation in a developing long bone is the epiphyseal plate (metaphysis), not the epiphysis itself [18]; (D) is false because the epiphyseal line is the inert, fused remnant of the epiphyseal plate after growth has ceased, so it is not responsible for active growth in length [12, 18]. Therefore, '6. None' (F) is the correct choice.

Question 10: All of the following statements are false, except?

- A. Metaphysis is a common site of infection osteomyelitis in children, because of poor blood supply.
- B. Defect in endochondral ossification affects growth and ossification of trunk.
- C. Growing end of the long bone is directed towards nutrient foramen, except fibula where nutrient foramen is opposite to growing bone.
- D. Secondary ossification form main part of the bone.
- E. None

Correct Answer: E. None

Explanation: All options are false: (A) is false because the metaphysis is a common site of osteomyelitis in children due to its sluggish, rich capillary loops, not poor blood supply; (B) is false because endochondral defects primarily affect the long limbs, leaving the trunk relatively spared; (C) is false because the growing end is directed away from the nutrient foramen (remembering: 'Towards the elbow I go, away from the knee I flee'); and (D) is false because the primary center (diaphysis) forms the main part of the bone. Therefore, 'None' of the options is true, making E the correct choice.

Question 11: What is the difference between compact bone and spongy bone?

- A. They have different bone marrow.
- B. They are made of different materials.
- C. They have different sizes of bone cells.
- D. They have different arrangement of bone cells.

Correct Answer: D. They have different arrangement of bone cells.

Explanation: Both compact and spongy (trabecular) bones are composed of the same cellular elements (osteocytes, osteoblasts, osteoclasts) and the same extracellular matrix materials (hydroxyapatite crystals and collagen fibers). However, they differ dramatically in their structural organization (arrangement): compact bone is dense and arranged into concentric lamellae called osteons, while spongy bone is arranged in a porous, branching network of spicules/trabeculae enclosing bone marrow [13].

Question 12: Which of the following is/are true about classification of bone based on shape?

- A. Clavicle and body of vertebra are modified type of long bone.
- B. Modified types of long bone have medullary cavity.
- C. Metacarpals, metatarsals and phalanges are typical long bones containing shaft and two epiphyses.
- D. Epiphyseal line indicates epiphyseal growth is just beginning.

Correct Answer: C. Metacarpals, metatarsals and phalanges are typical long bones containing shaft and two epiphyses.

Explanation: Long bones are tubular in shape and are characterized by a shaft (diaphysis) and two ends (epiphyses); this classification includes large limb bones as well as small bones like metacarpals, metatarsals, and phalanges [13]. Vertbrae are irregular bones [13]; modified long bones (such as the clavicle) lack a medullary cavity (A and B are false); and the epiphyseal line indicates that growth has ceased, not that it is beginning (D is false) [12, 18].

Question 13: Which of the following bone is not attached to any other bone in human body?

- A. Patella
- B. Hyoid bone
- C. Clavicle
- D. Malleus
- E. Incus

Correct Answer: B. Hyoid bone

Explanation: The hyoid bone, located in the anterior neck, is unique because it is the only bone in the human body that does not articulate directly with any other bone. It is held in place and suspended by muscles and ligaments. The patella (A) is a sesamoid bone that articulates with the femur; the clavicle (C) articulates with the sternum and scapula; and the malleus (D) and incus (E) are middle ear ossicles that articulate with each other.

Question 14: A thief was trying to escape from the police. He jumped from a window and fractured his femur and pelvis due to the force with which they were shoved into each other. Part of his femur bone broke through the skin. This fracture is:

- A. Depression
- B. Greenstick
- C. Spiral
- D. Comminuted
- E. Compacted

Correct Answer: E. Compacted

Explanation: An impacted or compacted fracture occurs when the broken ends of the bones are driven or shoved into each other by the force of the injury, as described here. A depression fracture (A) is typically seen in flat bones like the skull; a greenstick fracture (B) is an incomplete fracture in children [15]; a spiral fracture (C) is caused by a twisting force; and a comminuted fracture (D) is when the bone is splintered into multiple small fragments.

Question 15: Voluntary muscle is found at?

- A. Tongue
- B. Heart
- C. Stomach
- D. Uterine tube
- E. All

Correct Answer: A. Tongue

Explanation: Voluntary muscle is skeletal muscle. The tongue is composed of skeletal muscles (extrinsic and intrinsic), which are under voluntary control. In contrast, the heart consists of cardiac muscle (B) which is involuntary; and the stomach (C) and uterine tube (D) consist of smooth muscle, which is also involuntary.

Question 16: An entire skeletal muscle is surrounded by _____.

- A. Epimysium
- B. Perimysium
- C. Tendon sheath
- D. Sarcolemma
- E. Endomysium

Correct Answer: A. Epimysium

Explanation: Epimysium is the layer of dense connective tissue that surrounds the entire skeletal muscle. Perimysium (B) is the connective tissue layer that surrounds groups of muscle fibers called fascicles; endomysium (E) is the delicate connective tissue surrounding individual muscle fibers (cells); sarcolemma (D) is the cell membrane of a muscle cell; and a tendon sheath (C) is a fibrous sleeve surrounding a tendon.

Question 17: Which of the following statements about smooth muscle is true?

- A. Nuclei are peripherally located in the fibers.
- B. Smooth muscle is striated and involuntary.
- C. It has branching fibers.
- D. Fibers are small and spindle-shaped.
- E. Contractions are rapid and forceful.

Correct Answer: D. Fibers are small and spindle-shaped.

Explanation: Smooth muscle cells (fibers) are characterized by being small, fusiform (spindle-shaped), non-striated, and containing a single central nucleus. Skeletal muscle fibers have peripherally located nuclei (A is false); cardiac muscle is striated, involuntary (B is false of smooth), and has branching fibers (C is false); and smooth muscle contractions are typically slow, rhythmic, and sustained rather than rapid and forceful (E is false).

Question 18: Which muscle naming criteria are used to name the quadriceps femoris?

- A. Muscle action and location.
- B. The origin and insertion.
- C. Location and direction of muscle fibers.
- D. Location and number of origins.

Correct Answer: D. Location and number of origins.

Explanation: The name 'quadriceps femoris' is derived from: 'quadriceps' (meaning four heads, referring to the number of origins of the muscle) and 'femoris' (referring to the femur or thigh, which is its anatomical location). Thus, the muscle is named based on its location and number of origins.

Question 19: Which of the following muscles is NOT named after its location in the body?

- A. Deltoid
- B. Extensor carpi ulnaris
- C. Rectus abdominis
- D. Biceps femoris

Correct Answer: A. Deltoid

Explanation: The deltoid muscle is named after its shape (Greek letter delta, triangular). Extensor carpi ulnaris (B) is named for its action and location (ulnaris); rectus abdominis (C) is named for fiber direction (rectus) and location (abdominis); and biceps femoris (D) is named for its number of heads (biceps) and location (femoris).

Question 20: Which component of the cardiovascular system is involved in providing protection by acting as the body's second line of defense?

- A. Platelets
- B. Red blood cells
- C. White blood cells
- D. Formed elements
- E. Erythrocytes

Correct Answer: C. White blood cells

Explanation: White blood cells (leukocytes) are critical components of the immune system and represent the body's second line of defense (cellular immune response) against pathogens. Platelets (A) are involved in blood clotting; red blood cells/erythrocytes (B, E) transport oxygen; and formed elements (D) include all blood cells collectively.

Question 21: Which of the following is TRUE regarding the cardio-vascular system?

- A. Blood is the only stationary component of cardiovascular system.
- B. In venous drainage, blood always moves from the body tissues to the heart cavity.
- C. Coronary arteries are muscular arteries, so as ...
- D. Capillaries are the major determinates of systemic circuits.
- E. Musculo-venous pump is functionally known as peripheral heart.

Correct Answer: E. Musculo-venous pump is functionally known as peripheral heart.

Explanation: The musculo-venous pump refers to the mechanism whereby the contraction of skeletal muscles in the lower limbs compresses the deep veins, forcing blood upward toward the heart against gravity. Because this action is vital for maintaining venous return, it is clinically and functionally termed the 'peripheral heart' or 'venous heart'. Blood is mobile, not stationary (A is false); venous drainage can pass through portal systems before reaching the heart (B is false).

Question 22: Which of the following is TRUE regarding arterial anastomosis?

- A. Potential anastomosis refers to a communication of arterial trunks of nearly equal size.
- B. Actual anastomosis is a communication established by union of terminal arterioles.
- C. True end-artery is more likely serious than false end-anastomosis.
- D. In potential anastomosis, arterial communication takes place at capillary level.
- E. All of the above

Correct Answer: E. All of the above

Explanation: An anastomosis is a connection between two blood vessels. An 'actual anastomosis' (B) is a direct, active channel connecting terminal arterioles. A 'potential anastomosis' (A, D) represents collateral pathways that are normally closed or insignificant but can enlarge over time to maintain blood supply if a main vessel is blocked. A true end-artery occlusion is more serious than a false end-artery (where potential anastomoses exist; C is true). Thus, all statements are correct, making E ('All of the above') the correct answer.

Question 23: Oligodendrocytes:

- A. Serve the same function as islets of Langerhans cells.
- B. Serve the same function as microglial cells, but in the central nervous system.
- C. Serve the same function as microglial cells, but in the peripheral nervous system.
- D. Serve the same function as Schwann cells, but in the peripheral nervous system.
- E. Serve the same function as Schwann cells, but in the central nervous system.

Correct Answer: E. Serve the same function as Schwann cells, but in the central nervous system.

Explanation: Oligodendrocytes are glial cells responsible for myelinating axons in the Central Nervous System (CNS) [31]. Schwann cells perform the identical function of myelination but are located in the Peripheral Nervous System (PNS) [31]. Therefore, oligodendrocytes serve the same function as Schwann cells, but in the CNS.

Question 24: Which of the following is TRUE regarding neurons?

- A. Cell body is the metabolic center of neurons, almost identical to other types of cells.
- B. Dendrites are the only cytoplasmic extensions that project from the cell body.
- C. Multipolar neurons are the most common types of neurons in the CNS.
- D. Nodes of Ranvier indicate the myelinated regions of axons.
- E. Both sensory and motor neurons show ganglia outside the CNS.

Correct Answer: C. Multipolar neurons are the most common types of neurons in the CNS.

Explanation: Multipolar neurons, possessing a single axon and multiple dendrites, are the most common structural type of neuron in the central nervous system (CNS). Let's review why other choices are incorrect: Axons are also cytoplasmic extensions projecting from the soma (B is false); Nodes of Ranvier represent the unmyelinated gaps along an axon (D is false); and somatic motor neurons have their cell bodies inside the CNS (ventral horn of spinal cord), meaning they do not form motor ganglia outside the CNS (E is false).

Question 25: What are the two structural divisions of Peripheral Nervous System (PNS)?

- A. Somatic and autonomic
- B. Afferent and efferent
- C. Spinal and cranial nerves
- D. Sensory and motor
- E. Special and general

Correct Answer: C. Spinal and cranial nerves.

Explanation: The Peripheral Nervous System (PNS) is structurally divided into spinal nerves (which emerge from the spinal cord) and cranial nerves (which emerge from the brain) [12]. Somatic/autonomic, afferent/efferent, and sensory/motor are functional, rather than structural, classifications of the nervous system.

Question 26: Which of the following event is true regarding cleavage of zygote after 24 hours of fertilization?

- A. A series of special meiotic divisions resulting in blastomers.
- B. Production of highly differentiated blastomers.
- C. An increased cytoplasmic content of blastomers.
- D. A decrease in size and increase in number of blastomers.
- E. An increase in size and decrease in number of blastomers.

Correct Answer: D. A decrease in size and increase in number of blastomers.

Explanation: Cleavage consists of a rapid succession of mitotic (not meiotic) divisions where the zygote splits into smaller cells called blastomeres. Because there is no cell growth between these divisions (confined within the rigid shell of the zona pellucida), the total cytoplasmic volume does not increase, meaning that with each division, the blastomeres decrease in size while their number increases.

Question 27: Which of the following is incorrect regarding the phases of gametogenesis?

- A. Primordial germ cells originate from yolk sac and migrate to gonads.
- B. Increasing the numbers of germ cells by mitosis during migration.
- C. Reduction of chromosome number by meiosis during migration.
- D. Differentiation and maturation of gametes.
- E. Increasing numbers of germ cells by meiosis.

Correct Answer: C. Reduction of chromosome number by meiosis during migration.

Explanation: Primordial germ cells multiply exclusively by mitosis during their migration from the wall of the yolk sac to the developing gonadal ridges (A and B are true). Meiotic division (which reduces the chromosome number) does not begin until after migration is completed and the germ cells have settled into the gonads and initiated spermatogenesis or oogenesis. Therefore, statement C is incorrect and the correct answer.

Question 28: The correct order in ovarian cycle is respectively:

- A. Folliculogenesis – formation of corpus luteum – ovulation
- B. Formation of corpus luteum – ovulation – folliculogenesis
- C. Folliculogenesis – ovulation – formation of corpus luteum
- D. Formation of corpus luteum – ovulation – folliculogenesis.

Correct Answer: C. Folliculogenesis – ovulation – formation of corpus luteum

Explanation: The ovarian cycle proceeds sequentially in three major phases: (1) folliculogenesis, where the ovarian follicle grows and matures; (2) ovulation, where the follicle ruptures and releases the oocyte; and (3) the luteal phase, where the remaining follicular cells luteinize to form the corpus luteum.

Question 29: Which of the following statement is true regarding the final fate of somites?

- A. Differentiate into myotome which gives rise to muscle in trunk and limbs.
- B. Differentiate into sclerotome which gives rise to vertebrae and ribs.
- C. Arise from segmentation of paraxial mesoderm.
- D. Differentiate into dermatome which gives rise to dermis of skin.
- E. All of the above

Correct Answer: E. All of the above

Explanation: Somites are segmented blocks of paraxial mesoderm that form along the neural tube (C is true). Each somite differentiates into: (1) sclerotome, which forms the vertebrae and ribs (B is true); (2) myotome, which gives rise to the skeletal muscles of the trunk and limbs (A is true); and (3) dermatome, which forms the dermis of the skin (D is true). Thus, all statements are correct, making E the correct answer.

Question 30: Primitive streak enables identification of the:

- A. Craniocaudal axis
- B. Cranial from caudal
- C. Dorsal from ventral surface
- D. Right from left side
- E. All of the above

Correct Answer: E. All of the above

Explanation: The appearance of the primitive streak at the beginning of the third week of development is a landmark event that establishes all major body axes of the embryo: it runs along the craniocaudal axis (A); its primitive node is at the cranial end (B); it forms on the epiblast (dorsal surface) of the embryonic disc (C); and once the dorsal-ventral and craniocaudal axes are defined, the right and left sides are automatically established (D). Hence, E ('All of the above') is correct.

Question 31: Which of the following is/are not the derivatives of neural crest cells?

- A. Peripheral nervous system and their ganglia
- B. Chromaffin cells of the adrenal medulla
- C. Schwann cells
- D. Odontoblasts
- E. None of the above

Correct Answer: E. None of the above

Explanation: Neural crest cells are a highly multipotent population of cells that migrate extensively to give rise to a diverse array of structures, including: the sensory and autonomic neurons and ganglia of the PNS (A), the chromaffin cells of the adrenal medulla (B), Schwann cells responsible for myelinating peripheral nerves (C), and odontoblasts responsible for dentin formation in teeth (D). Because all of these are neural crest derivatives, 'None of the above' (E) is the correct answer.

Question 32: In measuring the length of the embryo which is the sitting height?

- A. Greatest length (GL)
- B. Crown-rump length (CRL)
- C. Crown-heel length (CHL)
- D. Carnegie embryonic staging system
- E. By counting the number of somites

Correct Answer: B. Crown-rump length (CRL)

Explanation: Crown-rump length (CRL) is the measurement of the length of the embryo from the top of the head (crown) to the bottom of the buttocks (rump), representing the 'sitting height'. Crown-heel length (CHL) is the 'standing height' from crown to heel, used for older fetuses. Greatest length (GL) (A) is used for very early pre-somatic embryos, and somite counting (E) is used during weeks 3-4 to estimate age.

Question 33: Derivative of the branchial arch nerves includes the following except?

- A. 1st branchial arch: Trigeminal (V) nerve including ophthalmic division
- B. 2nd branchial arch: Facial (VII) nerve
- C. 3rd branchial arch: Glossopharyngeal (IX) nerve
- D. 4th branchial arch: Vagus (X) nerve
- E. 6th branchial arch: Accessory (XI) nerve

Correct Answer: E. 6th branchial arch: Accessory (XI) nerve

Explanation: The nerves associated with the pharyngeal (branchial) arches are: 1st arch (Trigeminal, CN V), 2nd arch (Facial, CN VII), 3rd arch (Glossopharyngeal, CN IX), and 4th/6th arches (Vagus, CN X via superior laryngeal and recurrent laryngeal branches). The Accessory nerve (CN XI) is not a branchial arch nerve; it originates from spinal cord levels C1-C5 and innervates the sternocleidomastoid and trapezius muscles. Thus, E is the correct 'except' choice.

Question 34: Which embryonic cavity becomes the body cavity?

- A. Extraembryonic coelom
- B. Intraembryonic coelom
- C. Chorionic cavity
- D. Primitive yolk sac
- E. Exocoelomic cavity

Correct Answer: B. Intraembryonic coelom

Explanation: The intraembryonic coelom is a horseshoe-shaped cavity that develops within the lateral plate mesoderm. This internal cavity will eventually partition to form the three main body cavities: peritoneal, pleural, and pericardial cavities. The extraembryonic coelom (A), also known as the chorionic cavity (C) or exocoelomic cavity (E), is located outside the embryo proper and does not form the internal body cavities.

Question 35: Which of the following is false regarding embryonic folding?

- A. It happens during 4th week of development.
- B. It happens due to differential growth in different portions of embryo.
- C. Longitudinal folding results in bending of embryo so that the head and tail are brought together.
- D. During head folding the primitive heart and oropharynx move caudally and ventrally.
- E. All foregut, midgut and hind gut are formed by incorporation of yolk sac during longitudinal folding.

Correct Answer: E. All foregut, midgut and hind gut are formed by incorporation of yolk sac during longitudinal folding.

Explanation: During the fourth week of development, embryonic folding occurs in two planes: longitudinal (craniocaudal) and lateral (transverse). Longitudinal folding incorporates the yolk sac to form the foregut cranially and the hindgut caudally. However, the midgut is formed and enclosed primarily by the lateral folding of the embryonic body wall, which wraps around to seal the anterior abdominal wall (except at the vitelline duct). Therefore, statement E is false.

Question 36: Which of the following is false regarding human placenta?

- A. Fetal part is derived from chorionic frondosum.
- B. Maternal part is derived from decidua basalis.
- C. It is hemochorial discoid type.
- D. The layer of Nitabuch is found on chorionic plate of placenta.
- E. The placental membrane has only 2 layers during 1st trimester/full term.

Correct Answer: E. The placental membrane has only 2 layers during 1st trimester/full term.

Explanation: The placental membrane (barrier) is a structural partition that separates maternal and fetal blood. During the first trimester, it is thick and consists of 4 distinct layers: (1) syncytiotrophoblast, (2) cytotrophoblast, (3) extraembryonic mesoderm (connective tissue), and (4) endothelium of fetal capillaries. In the third trimester (full term), it thins to 2 layers (syncytiotrophoblast and endothelium) to maximize diffusion. Therefore, statement E is false.

Question 37: The right & left crus of diaphragm is developed from:

- A. Septum transversum
- B. Pleuroperitoneal membranes
- C. Dorsal mesentery of esophagus
- D. Mesoderm of body wall

Correct Answer: C. Dorsal mesentery of esophagus

Explanation: The diaphragm develops from four embryological sources: (1) the septum transversum (which becomes the central tendon), (2) the pleuroperitoneal membranes, (3) the dorsal mesentery of the esophagus (which becomes the crura of the diaphragm, enclosing the aorta/esophagus), and (4) the muscular ingrowth from the lateral body walls. Thus, the right and left crura develop from the dorsal mesentery of the esophagus.

Question 38: Which one is incorrectly matched regarding pharyngeal arches?

- A. First pouch: Auditory tube
- B. Fourth pouch: Ultimobranchial body
- C. 2nd to 4th membrane: disappear
- D. 4th and 6th arch: all laryngeal cartilages
- E. None

Correct Answer: E. None

Explanation: Let's review the matches: (A) First pouch forms the tympanic cavity and auditory (Eustachian) tube (correct); (B) Fourth/fifth pouch forms the ultimobranchial body, which gives rise to parafollicular C-cells of the thyroid gland (correct); (C) 2nd to 4th membranes/clefts disappear as the 2nd arch grows caudally to cover them, forming the temporary cervical sinus (correct); (D) 4th and 6th arches form all laryngeal cartilages (thyroid, cricoid, arytenoid, etc.) (correct). Since all matches are true, 'None' (E) is the correct answer.

Question 39: Which one of the following is incorrectly matched?

- A. Malformation: results from an intrinsically abnormal developmental process.
- B. Most of anomalies detected at five years of age.
- C. Teratogenic result at pre-embryonic time: major anomaly.
- D. Dysplasia: An abnormal organization of cells in tissue and its morphologic result.

Correct Answer: C. Teratogenic result at pre-embryonic time: major anomaly.

Explanation: The pre-embryonic period (first 2 weeks of development) is characterized by an 'all-or-none' response to teratogens: exposure either results in high mortality (spontaneous abortion) or the highly regulative cells compensate completely, resulting in normal development without malformations. Major congenital anomalies occur during the highly sensitive embryonic period (weeks 3 to 8) of active organogenesis. Thus, C is incorrectly matched.

Question 40: Hypoplasia of the mandible, cleft palate, micrognathia are features of:

- A. Treacher Collins syndrome (mandibulofacial dysostosis)
- B. Pierre Robin syndrome
- C. Di-George syndrome
- D. A and B are correct
- E. None

Correct Answer: D. A and B are correct

Explanation: Both Treacher Collins syndrome (A) and Pierre Robin syndrome (B) are classic first arch syndromes, caused by abnormal migration or proliferation of cranial neural crest cells into the first pharyngeal arch. Both are characterized by severe mandibular hypoplasia (micrognathia) and cleft palate. (In Pierre Robin, the micrognathia causes the tongue to fall backward, resulting in glossoptosis and a secondary cleft palate). Thus, both A and B are correct.

Question 41: Which one of the following is/are NOT CORRECT description about tissues?

- A. Tissues are formed by groups of closely related cells and their product known as intracellular matrix.
- B. Abundant microvilli which form prominent brush border on the luminal surface are characteristic of epithelial tissues that transport substances or particles on the surface.
- C. Transitional epithelium is a type of epithelium whose appearance varies considerably depending on the degree of distension to which it is subjected.
- D. Non-keratinized stratified squamous epithelium is normally found in the body regions where the surface is dry or subject to severe mechanical abrasion.
- E. Epithelial tissue is characterized by abundant extracellular matrix.

Correct Answer: E. Epithelial tissue is characterized by abundant extracellular matrix.

Explanation: Epithelial tissue is characterized by highly cellular sheets of tightly packed cells with very little extracellular (intracellular) matrix. In contrast, connective tissue is defined by having abundant extracellular matrix separating cells. Therefore, statement E is incorrect. Note that statement D is also technically incorrect because non-keratinized stratified squamous epithelium is found on wet surfaces (oral cavity, esophagus, vagina), whereas keratinized is found on dry surfaces (skin epidermis).

Question 42: Which of the following reasons makes histology dependent on the use of microscopes?

- A. The small size of cells and matrix components
- B. The small size of tissues
- C. The small size of organs
- D. All of the above
- E. None of the above

Correct Answer: A. The small size of cells and matrix components

Explanation: Histology (microscopic anatomy) is the study of cells and the extracellular matrix of tissues [71, 101]. Because individual cells and their sub-cellular components (such as organelles, fibers, and ground substance) are far below the resolution limit of the human eye (typically 0.1 mm), their visualization is entirely dependent on the magnifying power of light or electron microscopes. Organs (C) are large and visible to the naked eye (gross anatomy) [100].

Question 43: Which of the following body tissues is most frequently subject to wear and tear due to its superficial location?

- A. Nervous tissue
- B. Muscle tissue
- C. Connective tissue
- D. Epithelial tissue
- E. All of the above

Correct Answer: D. Epithelial tissue

Explanation: Epithelial tissue covers all external body surfaces (epidermis of the skin) and lines all internal cavities, tracts, and lumens of tubular organs (gastrointestinal, respiratory, and urogenital tracts). Because of this superficial, lining location, epithelial tissue is directly exposed to severe physical, mechanical, and chemical wear and tear, and is characterized by a rapid and constant rate of cell renewal and mitosis [48].

Question 44: Which of the following descriptions about epithelial tissues is wrong?

- A. Epithelial tissues mostly exist on the free body surfaces or line body cavities and lumina of tubular organs.
- B. Epithelial tissues have diverse embryological origin from all germ layers.
- C. The apical, lateral and basal surfaces of epithelial cells reveal structural modifications based on the functional requirements of the tissue.
- D. Epithelial tissues are avascular.
- E. All epithelial tissues rest on a basal lamina.

Correct Answer: E. All epithelial tissues rest on a basal lamina.

Explanation: Almost all epithelial tissues sit upon a basement membrane (composed of basal lamina and reticular lamina) which anchors them to the underlying connective tissue. However, there are specialized exceptions, such as the ependyma (glial lining of brain ventricles), some sensory neuroepithelia, or certain specialized endocrine cells that do not possess a formal basal lamina. All other choices are highly accurate facts about epithelia.

Question 45: In histology laboratory session, suppose you are given a histological slide containing all the four basic types of tissues. Which of the following histological features can help you to distinguish connective tissue from the other basic tissues?

- A. Highest cellularity
- B. Abundant cellular junctions connecting adjacent cells together
- C. Abundant intercellular substance separating the cells from each other
- D. The presence of blood vessels
- E. The presence of motor and sensory nerve endings

Correct Answer: C. Abundant intercellular substance separating the cells from each other

Explanation: The hallmark histological feature of connective tissue is its low cellularity and abundant extracellular (intercellular) matrix (fibers and ground substance) that separates the cells from one another. In contrast, epithelial, muscular, and nervous tissues have very high cellularity with cells packed closely together with minimal extracellular space [45]. Although connective tissue is vascular (D) and innervated (E), epithelium has higher cellularity (A, B).

Question 46: Tendons, ligaments, retinacula, cartilages or bones are chiefly composed of _____.

- A. Epithelial tissue
- B. Nervous tissues
- C. Muscle tissue
- D. Connective tissue
- E. All of the above

Correct Answer: D. Connective tissue

Explanation: Tendons, ligaments, retinacula, cartilage, and bone are all specialized forms of connective tissue [12, 13, 46]. Tendons and ligaments are dense regular connective tissue composed of parallel collagen fibers; cartilages (hyaline, elastic, fibrocartilage) are skeletal connective tissues; and bones are calcified, rigid skeletal connective tissues [12, 13].

Question 47: The basic tissue types that constitute the brain and spinal cord is _____.

- A. Connective tissue
- B. Skeletal tissue
- C. Blood tissue
- D. Nervous tissue
- E. Epithelium

Correct Answer: D. Nervous tissue

Explanation: The brain and spinal cord form the Central Nervous System (CNS), which is chiefly composed of nervous tissue [47, 50]. Nervous tissue consists of two major cell populations: neurons (which receive, process, and transmit electrical signals) and neuroglia (supporting cells such as astrocytes, oligodendrocytes, microglia, and ependymal cells) [31, 50].

Question 48: Most tissue undergo constant cell turnover because of continuous cell division and ongoing death of cells. The turnover rate of cells is very rapid and high in which type of the following tissue?

- A. Connective tissue
- B. Muscle tissue
- C. Epithelial tissue
- D. Nervous tissue

Correct Answer: C. Epithelial tissue

Explanation: Epithelial tissues exhibit extremely high rates of mitotic division and cell renewal because they line surfaces exposed to severe mechanical friction, acid, enzymes, and microbial attack (such as the outer epidermis of the skin, the lining of the stomach, and the intestinal mucosa) [48]. In contrast, connective, nervous, and muscular tissues have much slower cell turnover, with cardiac muscle and neurons having virtually no regenerative/mitotic capacity.

Question 49: Each biological tissue consists of two interacting structural components known as _____ and _____.

- A. Cells and ECM
- B. Soma and axon
- C. Epidermis and dermis
- D. Nucleus and cytoplasm

Correct Answer: A. Cells and ECM

Explanation: Biologically, every tissue is composed of two primary, interacting structural components: (1) cells (the living structural and functional units) and (2) the extracellular matrix (ECM) (the non-living product of the cells, consisting of fibers and ground substance, which provides mechanical and biochemical support) [49].

Question 50: Nerve tissues are principally composed two groups of cells: These are _____ and _____.

- A. Neurons and neuroglia
- B. Astrocytes and Schwann cells
- C. Soma and dendrites
- D. Myelinated and unmyelinated fibers

Correct Answer: A. Neurons and neuroglia

Explanation: Nervous tissue is composed of two major categories of cells: (1) Neurons, which are specialized for the transmission of electrical impulses, and (2) Neuroglia (or glial cells), which provide vital metabolic, mechanical, and protective support to the neurons [50].